# Numbers or Pixels? What Near-Perfect Rice Variety Scores Depend On

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TABLE I
GRAINS PER CLASS AND SPLIT AFTER CLEANING.

| Class | Train | Val | Test | Total |
|---|---|---|---|---|
| Arborio | 10500 | 2250 | 2250 | 15000 |
| Basmati | 10500 | 2250 | 2250 | 15000 |
| Ipsala | 10292 | 2205 | 2206 | 14703 |
| Jasmine | 10500 | 2250 | 2250 | 15000 |
| Karacadag | 10500 | 2250 | 2250 | 15000 |
| All | 52292 | 11205 | 11206 | 74703 |

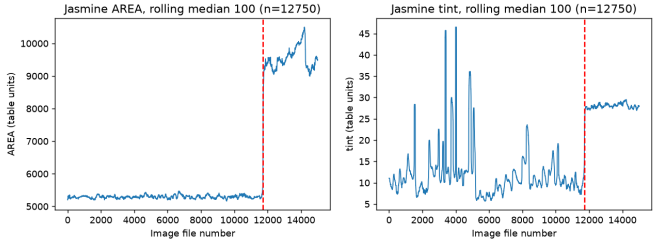



Fig. 1. Jasmine by file number: a break where the late lot begins.

***Abstract*—We ask whether measured numeric features or raw images classify five rice varieties better when both describe the identical grains. Both sit near the ceiling: a tabular gradient-boosted model reaches 99.88% test accuracy, a multilayer perceptron 99.92% (seed mean) and a small convolutional network 99.69% (seed mean). The ceiling hides a late Jasmine capture lot: models that have not seen it miss almost all of it, at high confidence. A small set of size-free features survives the same test. We report what the scores depend on, how our routing flags doubtful grains, and what the evidence does not show.**

## I. INTRODUCTION

Rice variety classification has been reported at near-perfect scores from images with convolutional networks [1] and from measured morphological and colour features [2]–[4]. The two literatures use different grains and different splits, so they cannot be compared directly. Our question is: do measured numeric features or raw images classify five rice varieties better on identical grains, and what do near-perfect scores depend on?

The report is organised as the brief requires: data and analysis, setup, results, a short discussion and a conclusion. We pursue three objectives. **O1:** compare model families on equal footing, with the same grains, the same splits, tuning on validation only, and one logged test read each. **O2:** explain the data behind the scores: row-to-image linking, duplicates, a late capture lot and clusters. **O3:** report what the scores do not show, through disclosures and limits.

## II. DATASET DESCRIPTION AND ANALYSIS

### A. Sources

The Rice Image Dataset holds 75000 images of five varieties (Arborio, Basmati, Ipsala, Jasmine, Karacadag). The Rice MSC dataset holds 106 numeric feature columns for the same grains: shape, colour statistics and texture measures. Table I gives the split sizes.

### B. Linking and cleaning

All 75000 links between table rows and images were rebuilt one-to-one. The median link distance is 0.001 for true links against 1.326 for shuffled links; 750 rows beyond the 99th percentile were kept. 297 exact duplicate pairs, all in Ipsala, were dropped (the later file of each pair), leaving 74703 grains. 8 rows have '?' cells and are median-imputed inside each model pipeline.

Two checks are reported but nothing is removed. Validity flags mark 10 columns (573266 cells in 63497 grain rows): the published entropy columns are negative because the published form is $-\sum s^2 \ln s^2$ over the non-zero pixel values of the whole image. We treat this as a quirk of the published formula, not as bad data. Robust-$z$ outliers (per class, $|z| > 6$) mark 71634 cells in 16181 rows and are kept.

### C. Exploratory analysis

Classes are balanced. The 15 `ALLdaub4` columns are near-duplicates of the mean colour columns, so we also use a `distinct91` set without them. Within Jasmine, a late lot of 3283 grains is larger and bluer than early Jasmine (Fig. 1): median length is $\times 1.35$, width $\times 1.32$, area $\times 1.81$, tint $\times 2.64$, and `EXTENT` $\times 1.15$. Its cause is not established. Late grains are spread across all three splits, so overall test accuracy does not measure this shift; the late-lot hold-out (Section IV-E) does.

### D. Feature selection

Mutual information, ANOVA and correlation-cluster pruning select $k = 16$ columns (validation accuracy 99.86%). We tried a grid of $k$ with HGB and logistic regression and took the smallest $k$ within one tenth of a point of each model's best validation accuracy, then the larger of the two. The authors' own top five

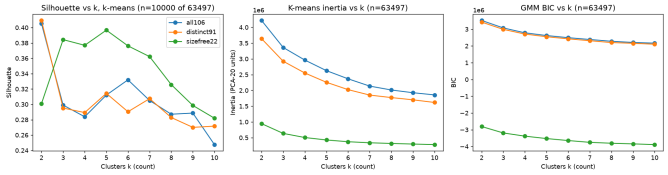



Fig. 2. Clustering against $k$: silhouette, $k$-means inertia and GMM BIC, for three feature sets.

shape features differ [2]: `ROUNDNESS`, `COMPACTNESS`, `SHAPEFACTOR_3`, `ASPECT_RATIO`, `ECCENTRICITY`. Our columns: `ROUNDNESS`, `SHAPEFACTOR_1`, `PERIMETER`, `MAJOR_AXIS`, `CONVEX_AREA`, `meanB`, `entropyL`, `meanA`, `StdDevV`, `StdDevL`, `entropyXX`, `meanS`, `meanH`, `StdDevCb`, `kurtosisYY`, `entropyH`.

### E. Clustering

As the unsupervised and generative view, we fit $k$-means and Gaussian mixtures (GMM) for $k$ from two to ten on a twenty-component PCA of all106 (cumulative variance 98.18%), train grains only, and score on validation with ARI, NMI, silhouette and BIC. The best GMM uses $k = 5$ (ARI 0.9437); the best $k$-means uses $k = 6$ (ARI 0.8934). On the size-free set the best GMM is lower (ARI 0.8052), again showing that size carries part of the variety signal (Fig. 2). The GMM silhouette at its best $k$ is 0.2952. At $k$ equal to six, $k$-means puts a share 0.9896 of the late lot into one cluster, of purity 0.8626: the unlabelled structure already separates the lot.

## III. Experimental Setup

Splits are stratified by class and lot, 70/15/15 into train, validation and test with seed zero (Table I). Hyperparameters are tuned on validation only. We then froze the code and a results hash, and made exactly 3 logged single test reads.

**Feature sets.** `all106`: all columns. `distinct91`: without the near-duplicates. `selected`: the pruned set above. `shape16`: shape columns. `colour90`: colour columns. `sizefree7`: seven shape ratios. `sizefree22`: those plus mean colour columns.

**Models.** Baselines: logistic regression, $k$-nearest neighbours, perceptron, naive Bayes, a Gaussian-mixture classifier, decision trees (including depth four) and histogram gradient boosting (HGB). Neural: an MLP on all106 and a small CNN (SmallCNN) on images, three seeds each. Routing chain: HGB answers when confident, the CNN takes grains HGB declines, and the rest are flagged unsure or unfamiliar.

**Neural settings.** The MLP takes all106, median-imputed and standardised on train; its best validation configuration has hidden layers of 256 and 128 units with ReLU, dropout 0.2 and learning rate 0.0003. The SmallCNN takes the whole 250 px crop resized uniformly to 128 px and scaled by the train channel mean and spread; it has four blocks of $3 \times 3$ convolution, batch normalisation, ReLU and $2 \times 2$ max-pooling, with base width 32 doubling per block, global average pooling, dropout 0 and a linear layer to five classes. Both use AdamW (weight decay $10^{-4}$, cosine schedule) and cross-entropy; the CNN learning rate is 0.001 and it uses horizontal and vertical flips only (no zoom, crop or colour jitter, so size and tint stay measurable). Table II lists the rest. The CNN trains on the Apple MPS GPU, the MLP on CPU; seeds are zero, one and two. Settings were tuned on validation, with the CNN grid run on a small train subset.

TABLE II
Training settings read from the training script.

| | MLP | SmallCNN |
|---|---|---|
| Batch size | 512 | 128 |
| Max epochs | 100 | 20 |
| Early stop (val. loss) | after 10 | after 4 |
| Input | 106 columns | 128 px RGB |
| Device | CPU | Apple MPS |

TABLE III
Test accuracy (%) by model and feature set.

| Model | all106 | selected | shape16 | colour90 |
|---|---|---|---|---|
| Logistic reg. | 99.83 | 99.78 | 97.73 | 98.95 |
| $k$NN | 99.79 | 99.83 | 97.91 | 98.43 |
| Perceptron | 99.67 | 99.54 | 96.44 | 97.45 |
| Naive Bayes | 99.05 | 99.08 | 97.18 | 94.91 |
| GMM classifier | 99.70 | 99.81 | 97.79 | 98.26 |
| Decision tree | 99.61 | 99.63 | 97.71 | 97.31 |
| Tree, depth four | 99.03 | — | — | — |
| HGB | 99.88 | 99.88 | 97.99 | 99.14 |
| MLP (seed mean) | 99.92 | — | — | — |

**Baseline settings.** Each baseline's grid point is chosen on validation macro-F1. For all106 the choices are: decision tree depth 20, $k$NN with $k = 3$, logistic regression $C = 100$, GMM classifier with 8 components, and HGB with learning rate 0.1, 150 iterations and 31 leaves. As a leakage guard we also ran each model on shuffled training labels: the best validation accuracy was 47.10% over 29 baseline rows (43.77% for the MLP, 14.48% for the CNN), and the stop rule (the ceiling, not a band around chance; see the disclosures) was not triggered.

**Metrics.** Accuracy and macro-F1 (classes are balanced, but per-class failure matters), Wilson intervals at the ninety-five percent level, and exact McNemar tests on paired predictions. The family fixed in advance is three pairs at seed zero (HGB against CNN as the primary pair, HGB against MLP, MLP against CNN) with Holm correction across the three; seeds one and two repeat the same three tests as a check.

## IV. Results

### A. Baselines

Table III gives test accuracy. Every model on all106 or selected is near the ceiling; shape alone and colour alone are lower (Fig. 3).

Table IV adds macro-F1, Wilson intervals and validation accuracy for the headline models. Test results agree with validation: each model is within a tenth of a point of its validation accuracy.

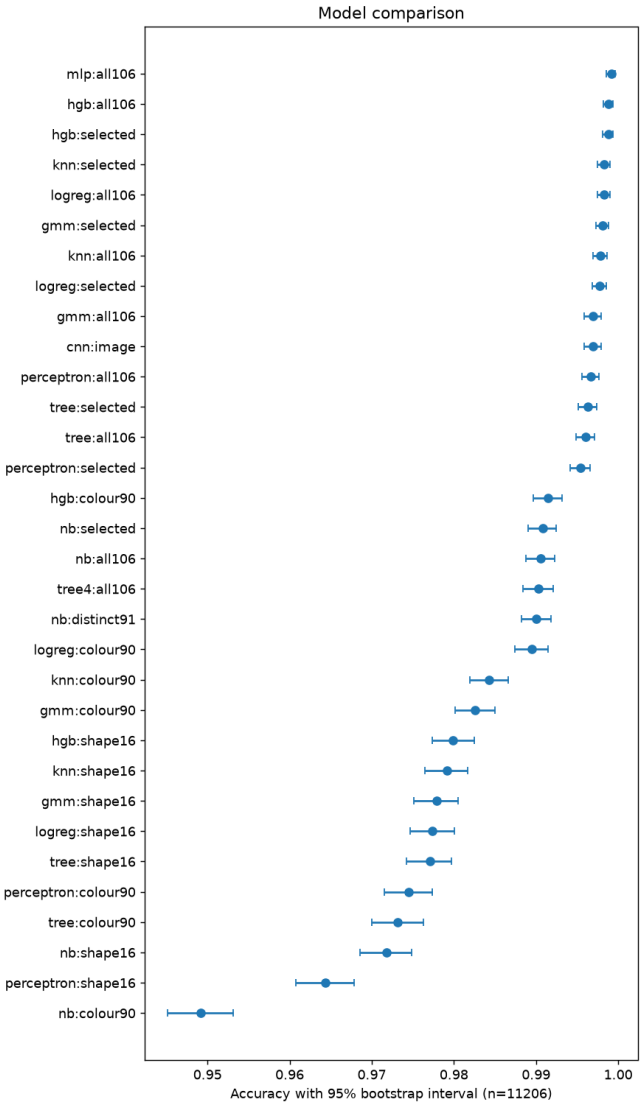



Fig. 3. Test accuracy across models and feature sets. Bars are bootstrap intervals; the tables use Wilson intervals.

TABLE IV
HEADLINE MODELS ON ALL106 (CNN ON IMAGES; MLP AND CNN ARE SEED MEANS): TEST ACCURACY WITH WILSON INTERVAL, TEST MACRO-F1 AND VALIDATION ACCURACY (ALL IN %).

| Model | Test acc. | Wilson interval | Test F1 | Val. acc. |
|---|---|---|---|---|
| HGB | 99.88 | 99.82–99.94 | 99.88 | 99.85 |
| MLP | 99.92 | 99.86–99.96 | 99.92 | 99.86 |
| CNN | 99.69 | 99.59–99.79 | 99.69 | 99.60 |
| Tree, depth four | 99.03 | 98.84–99.21 | 99.03 | 99.02 |

## B. Neural models

The MLP makes 11, 9 and 8 test errors on its three seeds (seed mean 99.92%, sd 0.01 points). The CNN makes between 15 and 50 errors (seed mean 99.69%, sd 0.16 points), so its seeds differ widely. Fig. 5 shows why this is plausible: the CNN validation loss jumps from epoch to epoch and early

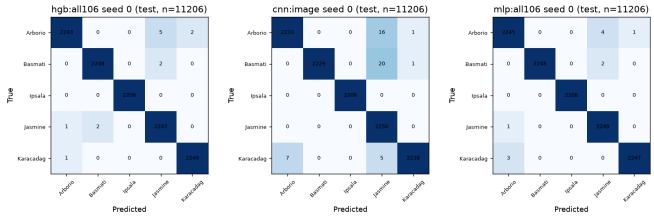



Fig. 4. Test confusion matrices of the compared models.

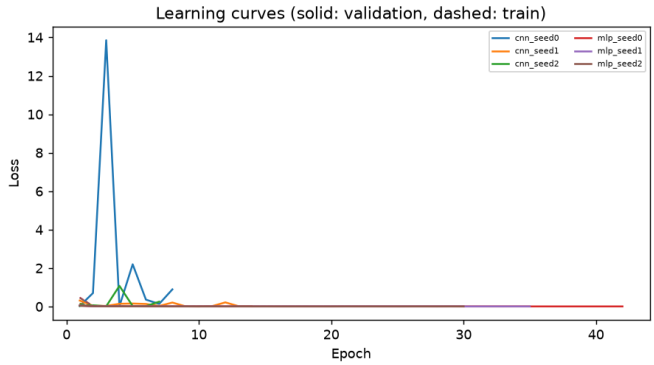



Fig. 5. Loss per epoch for the three MLP and three CNN seeds. The spikes are CNN validation loss.

stopping ends the three runs at different points, while the MLP curves are smooth. We do not claim this is the cause. Fig. 4 shows the test confusions.

## C. Paired tests

HGB against the MLP shows no significant difference on any seed (Holm-adjusted exact McNemar $p = 0.7266$, $0.7895$ and $0.2266$). Tabular against CNN differed significantly for seeds zero and two (Holm $p = 4.74 \times 10^{-7}$ and $1.38 \times 10^{-4}$) but not for seed one ($p = 0.8318$). The CNN gap therefore depends on the seed.

Table V gives the underlying counts. A pair's discordant grains are the only ones that carry information: in every HGB–MLP comparison the counts are small and balanced, whereas against the CNN the tabular model is right on many grains the CNN gets wrong for seeds zero and two.

## D. Routing

The routing chain works in three steps. HGB answers a grain when its confidence is above its floor; the CNN takes grains HGB declined, if its own confidence is above its floor; a table-side and an image-side familiarity gate flag grains unlike anything seen as unfamiliar. Grains neither model will answer are marked unsure. For seed zero, the chain answers a share 0.9797 of the 11206 test grains with 2 errors (accuracy 99.98%). The unsure share is 0.0046 and the unfamiliar share 0.0156. At a ninety-five percent coverage point the chain's accuracy is 99.99%. Floors were set in-sample on validation (see the disclosures), so these routing figures are optimistic.

TABLE V

PAIRED EXACT MCNEMAR TESTS ON THE TEST SPLIT: GRAINS ONLY THE FIRST MODEL GETS RIGHT, GRAINS ONLY THE SECOND GETS RIGHT, AND THE HOLM-ADJUSTED $p$. SEED ZERO AND TWO OF THE CNN COMPARISONS ARE THE SIGNIFICANT ONES AGAINST HGB AND MLP.

| Pair | Seed | First only | Second only | Holm $p$ |
|---|---|---|---|---|
| HGB vs MLP | zero | 3 | 5 | 0.7266 |
| HGB vs MLP | one | 3 | 7 | 0.7895 |
| HGB vs MLP | two | 3 | 8 | 0.2266 |
| HGB vs CNN | zero | 45 | 8 | $4.74 \times 10^{-7}$ |
| HGB vs CNN | one | 12 | 10 | 0.8318 |
| HGB vs CNN | two | 34 | 8 | $1.38 \times 10^{-4}$ |
| MLP vs CNN | zero | 46 | 7 | $1.20 \times 10^{-7}$ |
| MLP vs CNN | one | 13 | 7 | 0.7895 |
| MLP vs CNN | two | 36 | 5 | $2.35 \times 10^{-6}$ |

TABLE VI

LATE-LOT TEST RECALL (%): LOT HELD OUT VERSUS LOT IN TRAINING.

| Model | Held out | Mean conf. | In training |
|---|---|---|---|
| HGB all106 | 1.42 | 0.9263 | 100.00 |
| HGB shape16 | 0.00 | 0.9677 | 91.87 |
| HGB sizefree7 | 84.55 | 0.8990 | 92.48 |
| CNN seed zero | 0.41 | 0.8357 | 100.00 |
| CNN seed one | 1.42 | 0.8189 | 99.80 |
| CNN seed two | 16.06 | 0.6745 | 100.00 |

### *E. Late-lot hold-out*

We trained without the late Jasmine lot and tested on its late test grains ($n = 492$; Table VI). HGB on all106 recovers only 1.42% of them, at mean confidence 0.9263. The CNN seeds recover 0.41%, 1.42% and 16.06%. The size-free set sizefree7 keeps 84.55%. With the lot in training, recall returns.

### *F. Grey-world tint arm*

This arm asks whether the CNNs trained with the late lot lean on its blue tint. On the 492 late-lot test grains, grey-world colour correction (which removes the overall tint of an image) takes CNN recall from 99.93% (seed mean, uncorrected, same loading path) to 4.00%. So the corrected images are no longer recognised, but that alone does not show that colour is the reason. The pre-set check that the correction gain does not track grain area passed only for Arborio. For Jasmine the largest absolute correlation was 0.9293, so there colour cannot be separated from size.

## V. BRIEF DISCUSSION

**Decision trees in practice.** A depth-four tree reaches 99.03% test accuracy with rules a grading station can read and audit (Fig. 6). Its top splits use `SHAPEFACTOR_3`, then `MAJOR_AXIS` and `PERIMETER`: pixel-scale size measures. A change of camera scale, or a larger lot such as the late one, moves grains across such thresholds. A practical use is an interpretable first stage with a familiarity check, recalibrated per camera.

**Numbers versus pixels.** On identical grains the tabular models are at least as accurate as the CNN on every seed; the gap is significant for two of the three CNN seeds, and the MLP and HGB do not differ significantly. The near-perfect scores lean on size, which fails when the lot changes; size-free features trade a little accuracy for robustness.

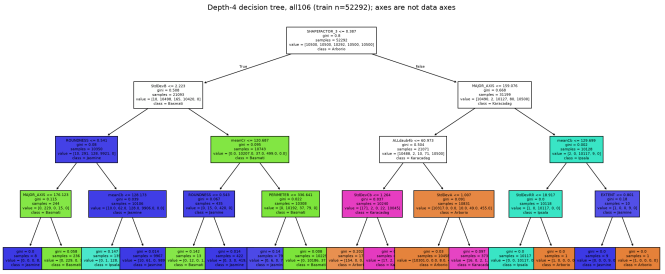



Fig. 6. Depth-four decision tree.

**What near-perfect means.** On 11206 test grains, HGB makes 13 errors, about 11.6 per ten thousand grains; the MLP seeds make 7.1 to 9.8, the CNN seeds 13.4 to 44.6, and the depth-four tree 97.3. HGB and the MLP differ by a handful of grains and their Wilson intervals in Table IV overlap; the depth-four tree is clearly lower, and the CNN depends on its seed. What matters more than the last decimal is whether a model's confidence is calibrated and whether it can say a grain is unfamiliar: the late lot was missed at high confidence, which no test accuracy reveals.

**Limitations.** One dataset, one source and camera; the cause of the late lot is not established; validation contains late grains; routing is calibrated in-sample; GPU nondeterminism makes seeds drift.

## VI. CONCLUSION

On identical grains, the tabular models are at least as accurate as the CNN on every seed, the gap is significant for two of three CNN seeds, and the MLP and HGB do not differ significantly. The near-perfect scores depend on size, and a late Jasmine lot that unseen models miss at high confidence exposes it, while size-free features survive. Future work: data from several lots, size normalisation, and an external camera.

## REFERENCES


[1] M. Koklu, I. Cinar, and Y. S. Taspinar, "Classification of rice varieties with deep learning methods," *Computers and Electronics in Agriculture*, vol. 187, p. 106285, 2021.

[2] I. Cinar and M. Koklu, "Determination of effective and specific physical features of rice varieties by computer vision in exterior quality inspection," *Selcuk Journal of Agricultural and Food Sciences*, vol. 35, no. 3, pp. 229–243, 2021.

[3] ——, "Identification of rice varieties using machine learning algorithms," *Journal of Agricultural Sciences (Tarim Bilimleri Dergisi)*, vol. 28, no. 2, pp. 307–325, 2022.

[4] ——, "Classification of rice varieties using artificial intelligence methods," *International Journal of Intelligent Systems and Applications in Engineering*, vol. 7, no. 3, pp. 188–194, 2019.

TABLE VII
DISCLOSURES: WHAT THE SCORES DO NOT SHOW.

| Item | One-line meaning | Evidence |
|---|---|---|
| old smoke overlap | the old smoke slice held real test grains; throwaway smoke models were fitted on them, no real model or choice used those models, and the slice now uses train and validation grains only | `results/smoke_overlap.csv` |
| in-sample calibration and routes | temperature, floors and cuts were set on the same validation rows | `results/routing/val_calibration.csv` |
| validation holds late grains | hyperparameters were picked on validation, which holds late-lot grains | `results/baselines/val_results.csv` |
| shuffled-label scores | a shuffled-label score can sit far from chance because a simple model can give a tight class one label, so the stop is the ceiling and not a band around chance | `results/baselines/control.csv` |
| overfit test limit | the first real overfit run missed its bar because too few updates fit in the planned epochs; the limit was raised after that failure, grains and bar unchanged | `results/run_log.csv` |
| cnn floor grains | the CNN floor was refit on the grains HGB declines, after a look at validation routes and before any test read; zero CNN validation errors is by construction | `results/code_lock.json` |
| eda counts include test | class counts in the EDA include test rows, as counts only | `results/eda/class_balance.csv` |
| image gate seed | the image familiarity gate uses seed zero for all three CNN seeds | `results/routing/val_routes.csv` |
| near-upright | the near-upright band is the group's own reading of the angle table | `results/compare/val_by_angle.csv` |
| perceptron scores | perceptron and kNN scores are not calibrated probabilities; they keep the predicted class only | `results/baselines/val_results.csv` |
| holdout cnn retrained | the hold-out CNNs were trained twice because an edit changed the reuse fingerprint; seed zero repeated exactly, seeds one and two drifted on the GPU and stopped earlier; the second set was frozen by procedure, not because it scored better, and no test read used the first set | `results/neural/epoch_times.csv` |
| tint gain vs area | the pre-set gain-versus-area check passed only for Arborio and failed outright for Jasmine, so for Jasmine the tint arm cannot separate colour from size | `results/robustness/greyworld_gain_vs_area.csv` |
| code changed after lock | one logged code change after the lock; the lock file records its time and reason | `results/code_lock.json` |

## APPENDIX A
## GROUP CONTRIBUTIONS

Brief §5: the contributions appendix is not counted in the page limit.

| Member | Stages / contribution | Confirmed |
|---|---|---|
| | to be confirmed by the group | |
| | to be confirmed by the group | |
| | to be confirmed by the group | |
| | to be confirmed by the group | |
| | to be confirmed by the group | |

Code and prompts: repository URL https://github.com/yahiaelmahdi7-bit/f20dl-rice-classification; code-generation prompts in `prompts/`.